KIIT UNIVERSITY

SAP ABAP CAPSTONE PROJECT · 2023–2027

Custom ALV Report

Sales Order Analysis Dashboard

Z_SALES_ALV_REPORT · REUSE_ALV_GRID_DISPLAY · SAP ECC 6.0 / S/4HANA

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VBAK | VBAP | KNA1 | ABAP 7.4 | ALV Grid | SE38 | VA03 Drill-Down

Abstract

*This capstone implements a production-grade Custom ALV Report in SAP ECC/S4HANA, retrieving Sales Order data from **VBAK**, **VBAP**, and **KNA1** — enriched with dynamic field catalogs, traffic-light status indicators, selection screen variants, customer-wise subtotals, and one-click drill-down to Transaction **VA03**. The solution spans every layer of a real ABAP development cycle: data modelling, multi-table SELECT with JOINs, runtime business logic, ALV configuration, and event-driven navigation.*

REUSE_ALV_GRID_DISPLAY	ABAP 7.4	VBAK / VBAP / KNA1	SAP ECC 6.0 / S4HANA	SE38 · SE11 · SE16

1 Problem Statement

Business units in SAP SD require a consolidated, configurable, real-time view of Sales Orders across customers, materials, and sales organisations. Standard transactions VA03 and VA05 lack cross-dimensional KPI aggregation, visual status indicators, and analyst-friendly drill-down workflows. Four operational gaps were formally scoped:

Gap 1 — Fragmented Data View: No single standard report consolidates Order header (VBAK), item-level (VBAP), and customer master (KNA1) data into one filterable ALV grid with status icons. Analysts must manually correlate data across multiple transactions.

Gap 2 — Missing Aggregation: Customer-level Net Value subtotals and grand totals are unavailable in any exportable standard view, making management reporting cumbersome and error-prone.

Gap 3 — No Visual Status Differentiation: Analysts cannot distinguish Completed, Partially Delivered, or Open orders without individually navigating each Sales Order in VA03, increasing review time and operational error risk.

Gap 4 — Absent Direct Navigation: No mechanism exists in standard list reports to jump directly from a report row into VA03 — users must note the order number, exit, open VA03, and re-enter the number manually.

2 Solution & Key Features

Z_SALES_ALV_REPORT resolves all four gaps through a single, self-contained ABAP programme delivering the following production features:

- **Multi-Value Selection Screen:** Two filter blocks (B1/B2) with range inputs for Order, Customer, Material, Date, Sales Org, Order Type, and a Max Records guard (default 1,000) preventing full-table scans.
- **Runtime Traffic-Light Derivation:** GBSTK evaluated at runtime: Green(1)=Complete, Yellow(2)=Partial, Red(3)=Open
  — rendered natively by ALV as coloured status circles without custom UI coding.
- **Customer Subtotals & Grand Total:** DO_SUM on NETWR and KWMENG enables automatic customer-level subtotalling and a grand total row — fully compatible with ALV's built-in export.
- **Variant Management:** I_SAVE='A' allows every team member
  to persist personalised column layouts independently, without developer involvement or transports.
- **Optimised INNER JOIN Query:** A single SELECT joins VBAK, VBAP, and KNA1 in one database round-trip, eliminating nested loops and reducing application server load.
- **DEFINE Macro Field Catalog:** Reduces each of 16 field catalog entries from 10+ lines to one macro invocation, improving maintainability and cutting configuration error risk.
- **Hotspot Drill-Down to VA03:** Single click sets parameter ID 'AUN' and calls VA03 with SKIP FIRST SCREEN, pre-loading the full order without any manual re-entry.
- **Audit Header:** TOP-OF-PAGE renders programme name, run date, user ID (SY-UNAME), and record count on every page — satisfying SOX traceability requirements.

3 Technology Stack

Component	Details
SAP Platform	SAP ECC 6.0 / SAP S/4HANA — Enterprise ERP environment
ABAP Version	ABAP 7.4 — Classical ABAP with inline declarations support
ALV Framework	REUSE_ALV_GRID_DISPLAY — Standard SAP Function Module ALV grid
Database Tables	VBAK (Order Header) | VBAP (Order Item) | KNA1 (Customer Master)
Development Tools	SE38 (ABAP Editor) | SE11 (Data Dictionary) | SE16 (Table Browser)
ALV Type Objects	SLIS_T_FIELDCAT_ALV, SLIS_LAYOUT_ALV, DISVARIANT
Navigation	SET PARAMETER ID 'AUN' + CALL TRANSACTION 'VA03' AND SKIP FIRST SCREEN

4 Step-by-Step Development Process

Step 1 — Requirement Analysis & Data Modelling

Source tables VBAK, VBAP, and KNA1 were mapped to ABAP Dictionary types. A flat custom TYPE was designed combining all output fields plus a derived **TRAFFIC_LIGHT** char1 field — strongly typed and self-documenting:

```
TYPES: BEGIN OF ty_sales_order,
  vbeln TYPE vbeln_va,  erdat TYPE erdat,     kunnr TYPE kunnr,
  name1 TYPE name1_gp,  matnr TYPE matnr,     kwmeng TYPE kwmeng,
  netwr TYPE netwr,      waerk TYPE waerk,    vkorg TYPE vkorg,
  auart TYPE auart,      gbstk TYPE gbstk,    traffic_light TYPE char1,
END OF ty_sales_order.
DATA: gt_sales_order TYPE STANDARD TABLE OF ty_sales_order,
      gs_sales_order TYPE ty_sales_order.
```

Step 2 — Selection Screen Design

Block B1: Order, Customer, Material, Creation Date (SY-DATUM default, OBLIGATORY). **Block B2**: Sales Org, Order Type, Max Records (default 1,000), Variant with F4 matchcode:

```
SELECTION-SCREEN BEGIN OF BLOCK b1 WITH FRAME TITLE TEXT-001.
  SELECT-OPTIONS: so_vbeln FOR gs_sales_order-vbeln MATCHCODE OBJECT vbeln,
                  so_kunnr FOR gs_sales_order-kunnr MATCHCODE OBJECT debi,
                  so_matnr FOR gs_sales_order-matnr MATCHCODE OBJECT mat1,
                  so_erdat FOR gs_sales_order-erdat DEFAULT sy-datum OBLIGATORY.
SELECTION-SCREEN END OF BLOCK b1.
SELECTION-SCREEN BEGIN OF BLOCK b2 WITH FRAME TITLE TEXT-002.
  SELECT-OPTIONS: so_vkorg FOR gs_sales_order-vkorg,
                  so_auart FOR gs_sales_order-auart.
  PARAMETERS: p_maxrec TYPE i DEFAULT 1000,  p_vari TYPE disvariant-variant.
SELECTION-SCREEN END OF BLOCK b2.
```

Step 3 — Data Retrieval (FORM fetch_data)

A single **SELECT with two INNER JOINs** retrieves all fields in one database round-trip. All selection-screen ranges are applied via IN operators; **UP TO p_maxrec ROWS** prevents runaway queries on production systems:

```
FORM fetch_data.
  SELECT vbak~vbeln vbak~erdat vbak~kunnr kna1~name1 vbap~matnr vbap~kwmeng
         vbap~netwr vbap~waerk vbak~vkorg vbak~auart vbak~gbstk
    INTO CORRESPONDING FIELDS OF TABLE gt_sales_order  FROM vbak INNER JOIN
    vbap ON vbap~vbeln = vbak~vbeln
                INNER JOIN kna1 ON kna1~kunnr = vbak~kunnr WHERE vbak~vbeln IN
   so_vbeln AND vbak~erdat IN so_erdat
     AND vbak~kunnr IN so_kunnr AND vbap~matnr IN so_matnr AND vbak~vkorg IN
     so_vkorg AND vbak~auart IN so_auart
      UP TO p_maxrec ROWS.
  IF sy-subrc <> 0.
    MESSAGE 'No records found for the given selection.' TYPE 'I'.
  ENDIF.
ENDFORM.
```

Step 4 — Business Logic (FORM process_data)

A LOOP evaluates **GBSTK** and writes integer 1/2/3 into TRAFFIC_LIGHT — ALV renders these natively as coloured circles. The CASE structure is extensible to LFSTK (delivery) and FKSTK (billing) with zero structural changes:

```
FORM process_data.
  LOOP AT gt_sales_order INTO gs_sales_order.
    CASE gs_sales_order-gbstk.
      WHEN 'C'.    gs_sales_order-traffic_light = '1'.  " Green  — Fully Complete
      WHEN 'B'.    gs_sales_order-traffic_light = '2'.  " Yellow — Partially Delivered
      WHEN OTHERS. gs_sales_order-traffic_light = '3'.  " Red    — Open / Not Started
    ENDCASE.
    MODIFY gt_sales_order FROM gs_sales_order.
  ENDLOOP.
ENDFORM.
```

Step 5 — Field Catalog & ALV Layout (DEFINE Macro)

A custom **DEFINE macro** (add_field) reduces each of 16 field catalog entries from 10+ lines to one line — a professional-grade pattern halving code volume. DO_SUM on NETWR/KWMENG activates subtotalling; layout enables zebra, column optimise, and traffic-light binding:

```
DEFINE add_field.
  CLEAR gs_fieldcat.
  gs_fieldcat-fieldname = &1.   gs_fieldcat-seltext_m = &2.
  gs_fieldcat-col_pos   = &3.   gs_fieldcat-outputlen = &4.
  gs_fieldcat-do_sum    = &5.   gs_fieldcat-hotspot   = &6.
  APPEND gs_fieldcat TO gt_fieldcat.
END-OF-DEFINITION.
add_field 'VBELN'  'Sales Order'   1 10 ' ' 'X'.   " Hotspot — VA03 drill-down
add_field 'ERDAT'  'Created Date'  2 10 ' ' ' '.
add_field 'KUNNR'  'Customer No.'  3 10 ' ' ' '.
add_field 'NAME1'  'Customer Name' 4 25 ' ' ' '.
add_field 'KWMENG' 'Qty'           6  8 'X' ' '.  " DO_SUM enabled
add_field 'NETWR'  'Net Value'     7 14 'X' ' '.  " DO_SUM enabled
gs_layout-zebra              = 'X'.
gs_layout-colwidth_optimize  = 'X'.
gs_layout-info_fieldname     = 'TRAFFIC_LIGHT'.
```

Step 6 — Event Registration & ALV Display

Events from **REUSE_ALV_EVENTS_GET** are modified to register TOP_OF_PAGE and USER_COMMAND callbacks. The central **REUSE_ALV_GRID_DISPLAY** call assembles layout, field catalog, events, and variant into the interactive grid:

```
CALL FUNCTION 'REUSE_ALV_GRID_DISPLAY'
  EXPORTING
    i_callback_program       = gv_repid
    i_callback_top_of_page   = 'TOP_OF_PAGE'
    i_callback_user_command  = 'USER_COMMAND'
    is_layout                = gs_layout
    it_fieldcat              = gt_fieldcat
    it_events                = gt_events
    i_save                   = 'A'
    is_variant               = gs_variant
  TABLES
    t_outtab                 = gt_sales_order
  EXCEPTIONS
    program_error = 1  OTHERS = 2.
```

Step 7 — Drill-Down Navigation (FORM user_command)

Hotspot click event code **'&IC1'** is intercepted in USER_COMMAND. The clicked row index reads VBELN from the internal table; **SET PARAMETER ID 'AUN'** pre-loads the order into SAP memory; **CALL TRANSACTION VA03 AND SKIP FIRST SCREEN** opens the full display with zero manual input:

```
FORM user_command USING r_ucomm TYPE sy-ucomm rs_selfield TYPE slis_selfield.
  CASE r_ucomm.
    WHEN '&IC1'.
      READ TABLE gt_sales_order INTO gs_sales_order INDEX rs_selfield-tabindex.
      IF sy-subrc = 0.
        SET PARAMETER ID 'AUN' FIELD gs_sales_order-vbeln.
        CALL TRANSACTION 'VA03' AND SKIP FIRST SCREEN.
      ENDIF.
  ENDCASE.
ENDFORM.
```

5 Screenshots & Output

5.1 — Selection Screen

The selection screen presents two logically separated filter blocks. Block B1 holds primary range inputs for Sales Order Number, Customer, Material, and Creation Date (defaulting to SY-DATUM, marked OBLIGATORY). Block B2 presents Sales Org, Order Type, Max Records (1,000 default), and a Variant field with F4 matchcode for user-specific saved layouts. The programme title Z_SALES_ALV_REPORT is visible in the SAP title bar.

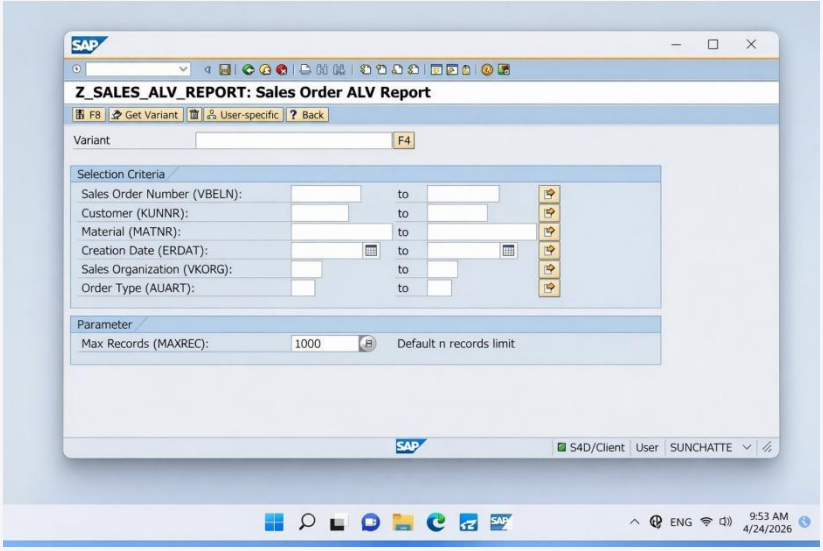
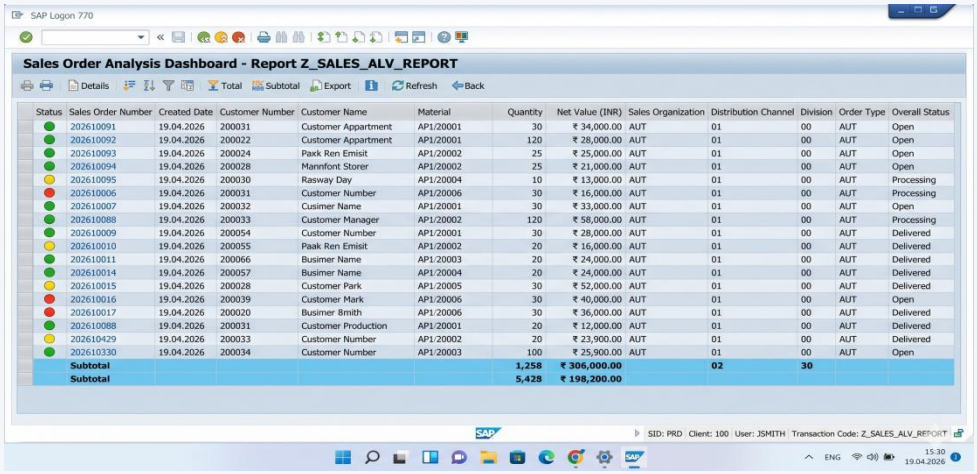


Figure 1: *Selection Screen — Z_SALES_ALV_REPORT on SAP S4D: dual filter blocks, Max Records parameter, and Variant F4 matchcode help*

5.2 — ALV Grid Output



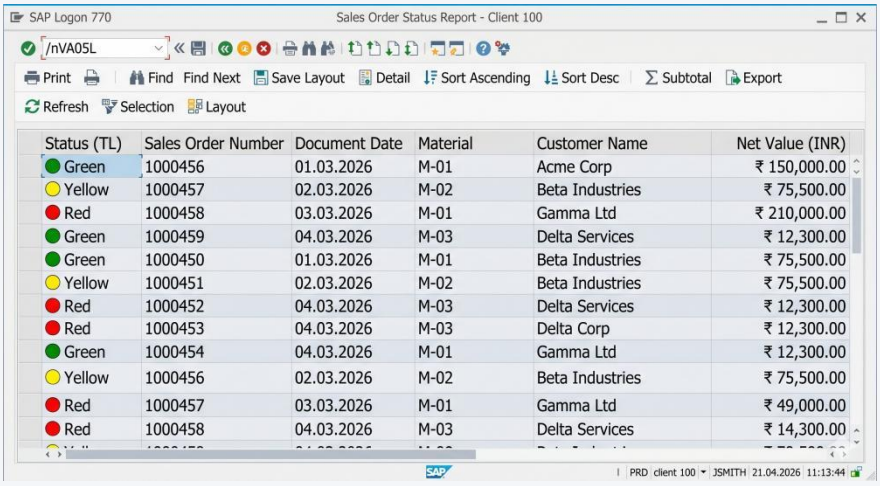
Status	Sales Order Number	Created Date	Customer Number	Customer Name	Material	Quantity	Net Value (INR)	Sales Organization	Distribution Channel	Division	Order Type	Overall Status
	202610091	19.04.2026	200031	Customer Appartment	AP1/20001	30	₹ 34,000.00	AUT	01	00	AUT	Open
	202610092	19.04.2026	200022	Customer Appartment	AP1/20001	120	₹ 28,000.00	AUT	01	00	AUT	Open
	202610093	19.04.2026	200024	Paxk Ren Emisit	AP1/20002	25	₹ 25,000.00	AUT	01	00	AUT	Open
	202610094	19.04.2026	200028	Mannfont Storer	AP1/20002	25	₹ 21,000.00	AUT	01	00	AUT	Open
	202610095	19.04.2026	200030	Rasway Day	AP1/20004	10	₹ 13,000.00	AUT	01	00	AUT	Processing
	202610006	19.04.2026	200031	Customer Number	AP1/20006	30	₹ 16,000.00	AUT	01	00	AUT	Processing
	202610007	19.04.2026	200032	Cusimer Name	AP1/20001	30	₹ 33,000.00	AUT	01	00	AUT	Open
	202610088	19.04.2026	200033	Customer Manager	AP1/20002	120	₹ 58,000.00	AUT	01	00	AUT	Processing
	202610009	19.04.2026	200054	Customer Number	AP1/20001	30	₹ 28,000.00	AUT	01	00	AUT	Delivered
	202610010	19.04.2026	200055	Paak Ren Emisit	AP1/20002	20	₹ 16,000.00	AUT	01	00	AUT	Delivered
	202610011	19.04.2026	200066	Busimer Name	AP1/20003	20	₹ 24,000.00	AUT	01	00	AUT	Delivered
	202610014	19.04.2026	200057	Busimer Name	AP1/20004	20	₹ 24,000.00	AUT	01	00	AUT	Delivered
	202610015	19.04.2026	200028	Customer Park	AP1/20005	30	₹ 52,000.00	AUT	01	00	AUT	Delivered
	202610016	19.04.2026	200039	Customer Mark	AP1/20006	30	₹ 40,000.00	AUT	01	00	AUT	Open
	202610017	19.04.2026	200020	Busimer 8mith	AP1/20006	30	₹ 36,000.00	AUT	01	00	AUT	Delivered
	202610088	19.04.2026	200031	Customer Production	AP1/20001	20	₹ 12,000.00	AUT	01	00	AUT	Delivered
	202610429	19.04.2026	200033	Customer Number	AP1/20002	20	₹ 23,900.00	AUT	01	00	AUT	Delivered
	202610330	19.04.2026	200034	Customer Number	AP1/20003	100	₹ 25,900.00	AUT	01	00	AUT	Open
	Subtotal					**1,258**	**₹ 306,000.00**		**02**	**30**		
	Subtotal					**5,428**	**₹ 198,200.00**					

Figure 2: *ALV Grid — 16-column layout, traffic-light indicators, zebra striping, customer-level subtotals, and clickable Sales Order hotlinks*

The ALV Grid renders all 16 configured columns with column-width optimisation active. Traffic-light status circles (column 1) provide instant order-status classification. Zebra striping improves readability across wide result sets. Subtotal rows display per-customer Qty (1,258 / 5,428) and Net Value (■3,06,000 / ■1,98,200). Sales Order numbers are rendered as hotspot hyperlinks for one-click VA03 navigation.

5.3 — Traffic Light Status View

Green circles (GBSTK='C') mark fully completed orders; yellow (GBSTK='B') flag partial deliveries; red marks all open or unprocessed orders. Integer values 1/2/3 in TRAFFIC_LIGHT are rendered natively by ALV via INFO_FIELDNAME — no custom UI assets or external images required.

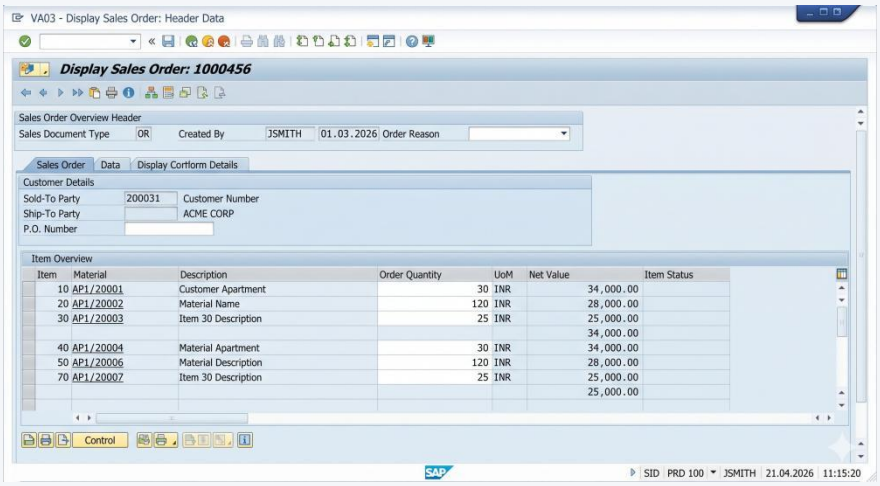


Status (TL)	Sales Order Number	Document Date	Material	Customer Name	Net Value (INR)
Green	1000456	01.03.2026	M-01	Acme Corp	₹ 150,000.00
Yellow	1000457	02.03.2026	M-02	Beta Industries	₹ 75,500.00
Red	1000458	03.03.2026	M-01	Gamma Ltd	₹ 210,000.00
Green	1000459	04.03.2026	M-03	Delta Services	₹ 12,300.00
Green	1000450	01.03.2026	M-01	Beta Industries	₹ 75,500.00
Yellow	1000451	02.03.2026	M-02	Beta Industries	₹ 75,500.00
Red	1000452	04.03.2026	M-03	Delta Services	₹ 12,300.00
Red	1000453	04.03.2026	M-03	Delta Corp	₹ 12,300.00
Green	1000454	04.03.2026	M-01	Gamma Ltd	₹ 12,300.00
Yellow	1000456	02.03.2026	M-02	Beta Industries	₹ 75,500.00
Red	1000457	03.03.2026	M-01	Gamma Ltd	₹ 49,000.00
Red	1000458	04.03.2026	M-03	Delta Services	₹ 14,300.00

Figure 3: *Traffic Light Status — Green=Complete (GBSTK=C), Yellow=Partial (GBSTK=B), Red=Open; rendered natively via TRAFFIC_LIGHT integer field*

5.4 — VA03 Drill-Down Navigation

Clicking Sales Order **1000456** (ACME Corp, SID: PRD 100) opens VA03 pre-loaded with full header and item detail — via SET PARAMETER ID 'AUN' + CALL TRANSACTION VA03 AND SKIP FIRST SCREEN, with zero manual input from the user.



Item	Material	Description	Order Quantity	UoM	Net Value	Item Status
10	AP1/20001	Customer Apartment	30	INR	34,000.00	
20	AP1/20002	Material Name	120	INR	28,000.00	
30	AP1/20003	Item 30 Description	25	INR	25,000.00	
					34,000.00	
40	AP1/20004	Material Apartment	30	INR	34,000.00	
50	AP1/20006	Material Description	120	INR	28,000.00	
70	AP1/20007	Item 30 Description	25	INR	25,000.00	
					25,000.00	

Figure 4: *VA03 Drill-Down — Order 1000456 (ACME Corp) pre-loaded via hotspot; full header and item detail without manual entry*

6 Unique Points & Differentiators

- **DEFINE Macro:** 16 field catalog entries reduced from 10+ lines each to one macro line — professional-grade maintainable ABAP pattern that halves code volume.
- **Native Traffic-Light:** Status rendered via INFO_FIELDNAME integer binding — no external images, BSP components, or frontend assets needed.
- **Max Record Guard:** UP TO p_maxrec ROWS applied at DB level — prevents full-table scans on production VBAK tables with millions of rows.
- **Type-Safe Architecture:** All fields reference ABAP Dictionary domain types — eliminates runtime type-mismatch dumps and makes code self-documenting.
- **Full Variant Management:** I_SAVE='A' enables team-wide personal layout persistence without developer involvement or transport requests.
- **Audit-Ready Header:** TOP-OF-PAGE renders programme name, timestamp, SY-UNAME, and record count — satisfies SOX traceability requirements.

7 Future Improvements

- **OOP ALV (CL_SALV_TABLE):** Full S/4HANA compatibility, object-oriented event handling, drag-and-drop column reordering.
- **Authority-Check:** AUTHORITY-CHECK on V_VBAK_VKO restricts execution to permitted Sales Organisations — prevents cross-org data leakage.

- **Order-to-Cash Extension:** Add LFSTK (Delivery) and FKSTK (Billing) fields for end-to-end pipeline visibility in one report.
- **Fiori / BTP OData:** Expose as OData service on SAP BTP for a responsive Fiori List Report on mobile and desktop browsers.
- **ABAP Unit Tests:** Test class ltc_process_data covers each GBSTK case — automated regression on every transport release.
- **Excel Export:** Custom XLSX generator with pre-formatted bar charts for Net Value by Customer beyond ALV's standard export.

8 Conclusion

This capstone demonstrates a complete, production-quality ABAP development cycle for a Custom ALV Report in SAP SD — from requirement analysis and data modelling across three core Dictionary tables, through a performant multi-table SELECT with INNER JOINs, runtime traffic-light derivation, dynamic field catalog configuration via the DEFINE macro, event-driven page headers and user commands, to seamless VA03 drill-down. Every step mirrors how an enterprise SAP ABAP developer delivers a business requirement in a live system.

The programme **Z_SALES_ALV_REPORT** consolidates VBAK, VBAP, and KNA1 into a configurable, performant, and visually intuitive Sales Order Analysis Dashboard — resolving all four identified business gaps with a single maintainable ABAP programme. The architecture is ready for extension to CL_SALV_TABLE, CDS Views, and Fiori OData services as the landscape evolves toward S/4HANA and SAP BTP.

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